ASSESSMENT TOOL-3 PART-3 (CO-3)

Data interpretation-PPO, TRPO & DDPG performance analysis.

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Course code: MLA0305

Course Name: REINFORCEMENT LEARNING

Slot: B

CODE:

```
import numpy as np

import matplotlib.pyplot as plt

# ==========================================

# PPO, TRPO & DDPG PERFORMANCE ANALYSIS

# ==========================================

algorithms = ["PPO", "TRPO", "DDPG"]

# ----------------------------

# Sample performance data

# ----------------------------

episodes = np.arange(1, 11)

ppo_reward = [80, 105, 130, 155, 175, 195, 215, 228, 238, 245]

trpo_reward = [70, 90, 112, 135, 155, 172, 188, 200, 210, 218]

ddpg_reward = [55, 72, 90, 108, 125, 140, 153, 165, 176, 185]

average_reward = [245, 218, 185]

success_rate = [92, 87, 81]

stability = [90, 85, 78]

# ==========================================

# GRAPH 1 - LEARNING CURVE

# ==========================================

plt.figure(figsize=(8,5))

plt.plot(episodes, ppo_reward,

        marker='o', label='PPO')
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plt.plot(episodes, trpo_reward,
         marker='s', label='TRPO')
plt.plot(episodes, ddpg_reward,
         marker='^', label='DDPG')
plt.xlabel("Episode")
plt.ylabel("Average Reward")
plt.title("Learning Curve of PPO, TRPO and DDPG")
plt.legend()
plt.grid(alpha=0.3)
plt.show()
# ==========================================
# GRAPH 2 - PERFORMANCE COMPARISON
# ==========================================
x = np.arange(len(algorithms))
width = 0.25
plt.figure(figsize=(8,5))
plt.bar(x - width,
        average_reward,
        width,
        label="Average Reward")
plt.bar(x,
        success_rate,
        width,
        label="Success Rate (%)")
plt.bar(x + width,
        stability,
        width,
        label="Stability (%)")
plt.xticks(x, algorithms)
plt.xlabel("Algorithm")
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plt.ylabel("Performance Value")
plt.title("Performance Comparison of PPO, TRPO and DDPG")
plt.legend()
plt.grid(axis='y', alpha=0.3)
plt.show()
# ==========================================
# GRAPH 3 - REWARD DISTRIBUTION
# ==========================================
ppo_scores = [220, 230, 238, 242, 245,
              248, 250, 252, 246, 240]
trpo_scores = [195, 205, 212, 216, 218,
               220, 223, 225, 215, 210]
ddpg_scores = [160, 170, 178, 182, 185,
               188, 190, 193, 180, 175]
plt.figure(figsize=(8,5))
plt.boxplot(
    [ppo_scores, trpo_scores, ddpg_scores],
    tick_labels=algorithms
)
plt.xlabel("Algorithm")
plt.ylabel("Reward")
plt.title("Reward Distribution Comparison")
plt.grid(axis='y', alpha=0.3)
plt.show()
# ==========================================
# SUMMARY RESULT
# ==========================================
print("\n")
print("=" * 55)
print("            SUMMARY RESULT")
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print("=" * 55)
print("\nAverage Reward:")
print("PPO  :", average_reward[0])
print("TRPO :", average_reward[1])
print("DDPG :", average_reward[2])
print("\nSuccess Rate:")
print("PPO  :", success_rate[0], "%")
print("TRPO :", success_rate[1], "%")
print("DDPG :", success_rate[2], "%")
print("\nStability:")
print("PPO  :", stability[0], "%")
print("TRPO :", stability[1], "%")
print("DDPG :", stability[2], "%")
# Find best algorithm
best_reward_index = np.argmax(average_reward)
best_success_index = np.argmax(success_rate)
best_stability_index = np.argmax(stability)
best_algorithm = algorithms[best_reward_index]
print("\n" + "=" * 55)
print("INTERPRETATION")
print("=" * 55)
print("\n1. PPO achieved the highest average reward of",
    average_reward[0])
print("2. TRPO achieved an average reward of",
    average_reward[1])
print("3. DDPG achieved an average reward of",
    average_reward[2])
print("4. PPO achieved the highest success rate of",
    success_rate[0], "%")
print("5. PPO achieved the highest stability of",
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    stability[0], "%")
print("\n" + "=" * 55)
print("FINAL RESULT")
print("=" * 55)
print("Best Performing Algorithm :", best_algorithm)
print("\nPPO performs better than TRPO and DDPG")
print("in terms of reward, success rate and stability.")
print("=" * 55)
```

SUMMARY OUTPUT:

```
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                SUMMARY RESULT
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Average Reward:
PPO   : 245
TRPO  : 218
DDPG  : 185

Success Rate:
PPO   : 92 %
TRPO  : 87 %
DDPG  : 81 %

Stability:
PPO   : 90 %
TRPO  : 85 %
DDPG  : 78 %

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INTERPRETATION
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1. PPO achieved the highest average reward of 245
2. TRPO achieved an average reward of 218
3. DDPG achieved an average reward of 185
4. PPO achieved the highest success rate of 92 %
5. PPO achieved the highest stability of 90 %

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FINAL RESULT
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Best Performing Algorithm : PPO

PPO performs better than TRPO and DDPG
in terms of reward, success rate and stability.
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```
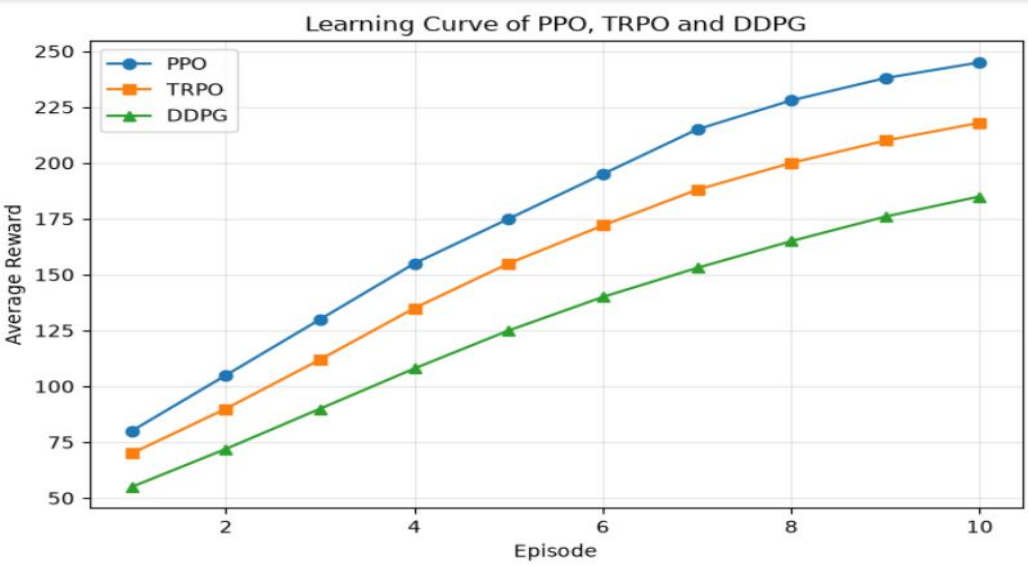
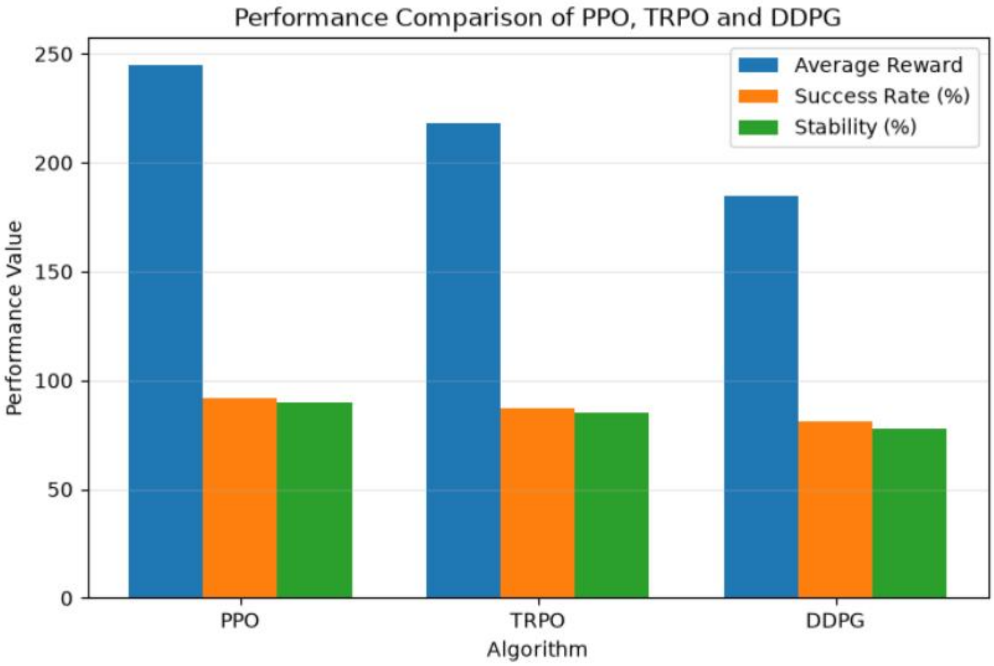
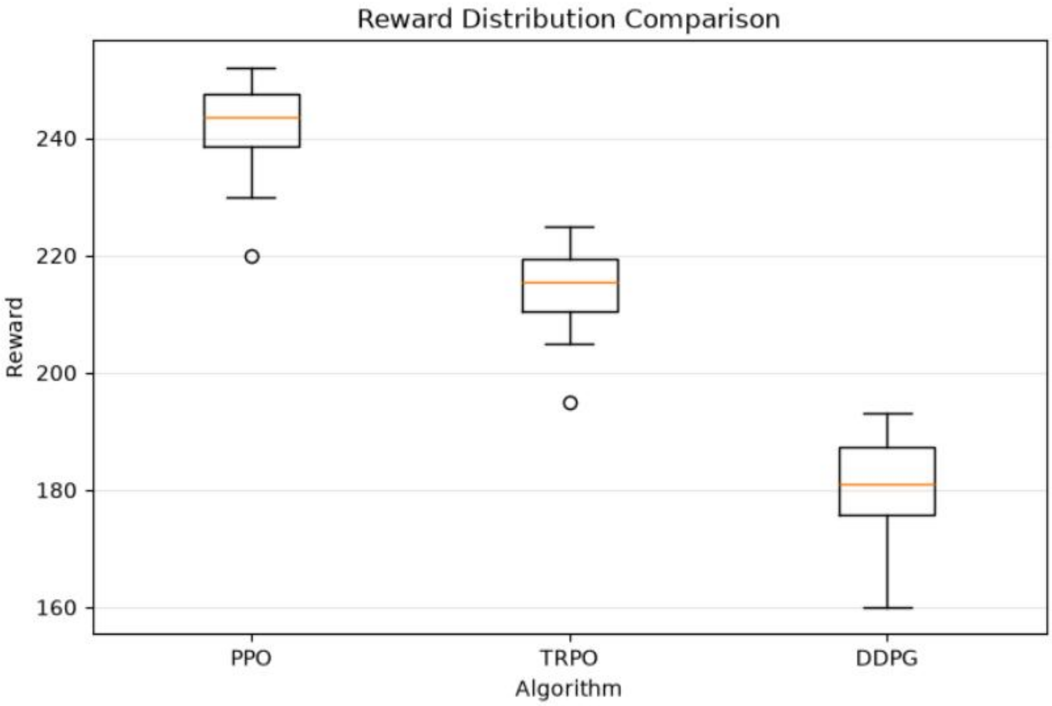

VISUALIZATION OUTPUT:




Performance Comparison of PPO, TRPO and DDPG
Average Reward
Success Rate (%)
Stability (%)
Performance Value
250
200
150
100
50
0
PPO
TRPO
DDPG
Algorithm


Reward Distribution Comparison
Reward
240
220
200
180
160
PPO
TRPO
DDPG
Algorithm